

BLAST Analysis of Escherichia coli DNA Ligase Protein.

For this analysis, a DNA ligase protein from Escherichia coli was selected. The selected sequence is a protein sequence with a length of 440 amino acids. The sequence was obtained from the NCBI Protein database. For sequence similarity analysis, the BLASTP (protein-protein BLAST) program was used. The selected protein sequence was compared with other protein sequences available in the database to identify highly similar sequences and study their sequence relationships.

☒ select all 100 sequences selected

[GenPept](#) [Graphics](#) [Distance tree of results](#) [Multiple alignment](#) [MSA Viewer](#)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	ATP-dependent DNA ligase [Escherichia coli]	Escherichia coli	906	906	100%	0.0	100.00%	440	WP_089689375.1
☒	DNA ligase [Escherichia coli]	Escherichia coli	906	906	100%	0.0	99.77%	440	HHU8355539.1
☒	DNA ligase [Escherichia coli]	Escherichia coli	905	905	100%	0.0	99.77%	440	MCS1056921.1
☒	DNA ligase [Escherichia coli]	Escherichia coli	905	905	100%	0.0	99.77%	440	EFM7744750.1
☒	ATP-dependent DNA ligase [Escherichia coli]	Escherichia coli	904	904	100%	0.0	99.77%	440	WP_447514889.1
☒	DNA ligase [Escherichia coli]	Escherichia coli	904	904	100%	0.0	99.77%	440	HBL7226550.1
☒	DNA ligase [Escherichia coli]	Escherichia coli	904	904	100%	0.0	99.77%	440	EEY7943599.1
☒	ATP-dependent DNA ligase [Escherichia coli]	Escherichia coli	904	904	100%	0.0	99.77%	440	WP_072647762.1
☒	DNA ligase [Escherichia coli]	Escherichia coli	904	904	100%	0.0	99.77%	440	HGF6116020.1
☒	DNA ligase [Escherichia coli]	Escherichia coli	904	904	100%	0.0	99.55%	440	EHS4348492.1

Figure 1. BLASTP search results showing significant matches of the selected DNA ligase protein.

The selected Escherichia coli DNA ligase protein sequence was analyzed using BLASTP. The BLAST search was performed to identify similar protein sequences available in the database.

The search produced several highly similar DNA ligase sequences. The top hits showed 100% query coverage and an E-value of 0.0. The percentage identity of the top matches was very high, ranging from 99.55% to 100%.

Pairwise Alignment Result

A highly similar BLAST hit was selected for detailed pairwise sequence alignment. The query and subject sequences showed a very high level of similarity.

The alignment length was 440 amino acids. Out of these, 439 amino acids were identical between the query and subject sequence .NO gaps were observed in the alignment.

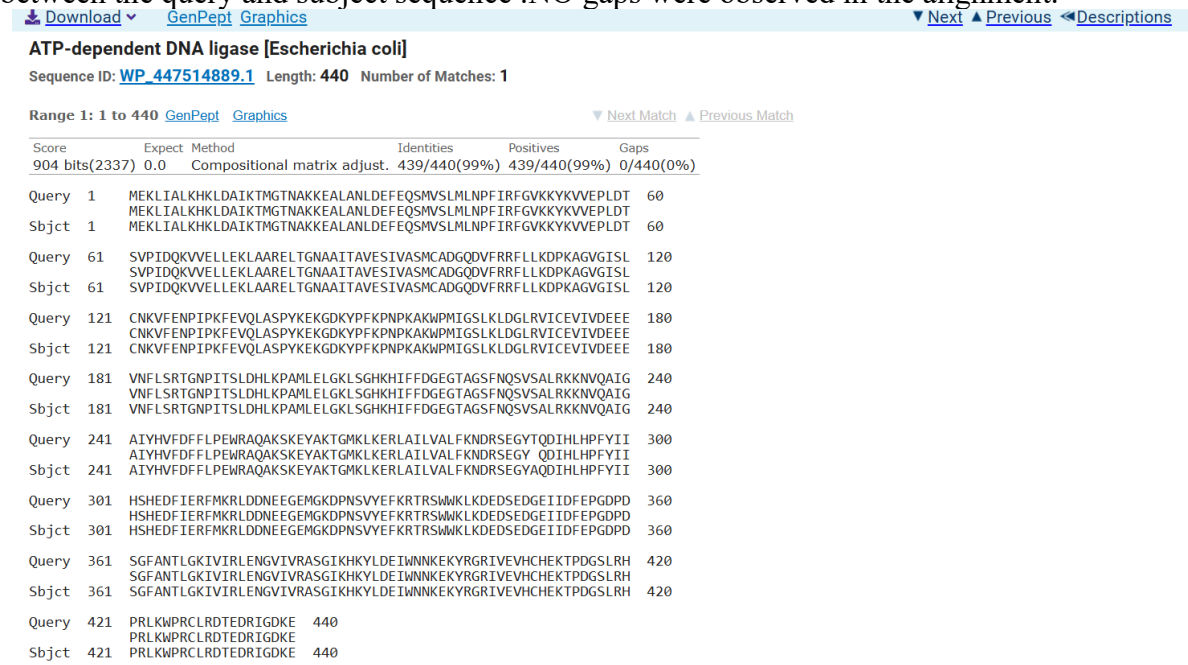


Figure 2. Pairwise alignment between the query DNA ligase protein and the selected subject sequence.

A highly similar BLAST hit was selected for detailed pairwise sequence alignment. The query and subject sequences showed a very high level of similarity.

The alignment length was 440 amino acids. Out of these, 439 amino acids were identical between the query and subject sequences .

Graphical Summary of BLAST Results

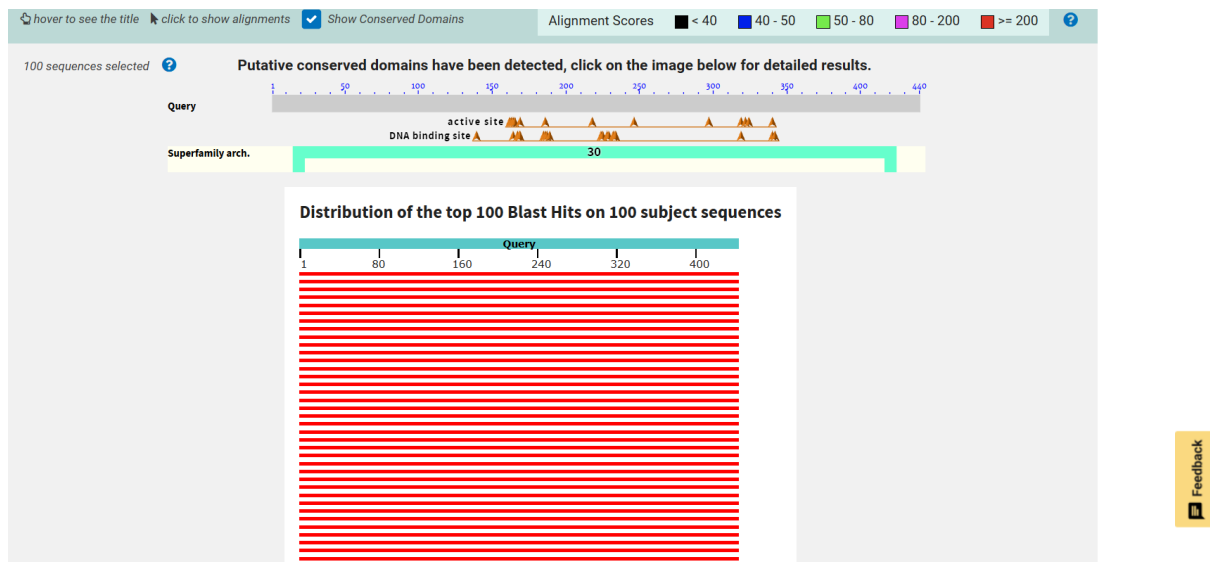


Figure: Distribution of the Top 100 BLAST Hits on 100 Subject Sequences.

Explanation:

The graphical summary represents the BLAST results obtained by comparing the *Escherichia coli* DNA ligase protein sequence with related protein sequences present in the database. The query sequence is approximately 420 amino acids in length, as shown by the scale on the horizontal axis. The horizontal bars represent the subject sequences that show similarity with the query protein.

The colour of each bar represents the alignment score of the corresponding BLAST hit. Red bars indicate alignment scores of ≥ 200 , which represent highly significant and strong sequence similarity. In this analysis, most of the BLAST hits are shown in red and extend across a large portion of the query sequence. This indicates that the query protein has high similarity with the identified subject sequences and that a major part of its sequence is conserved.

The upper portion of the figure also shows predicted functional and conserved regions of the protein. The active-site and DNA-binding-site markers indicate regions that may be important for the biological activity and interaction of the DNA ligase protein. The detected conserved domain, labelled as “Superfamily arch.”, further supports the presence of a conserved structural or functional region.